Noise-Adapted Tensor Network Maximum Likelihood Decoder for Spatially Varying, Hardware-Informed Noise

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Quantum error correction on near-term hardware must contend with spatially inhomogeneous noise that can be characterized experimentally. We implement a noise-adapted maximum likelihood (ML) decoder for the surface code using PEPS tensor-network contraction and benchmark it against minimum-weight perfect matching (MWPM) under three noise models: heterogeneous depolarizing noise, spin-qubit Z dephasing, and exchange-only qubit biaxial X, Z noise. The decoder is implemented in the python and is compatible with the `stim` stabilizer simulator. Across all models and code distances $d \in \{3, 5, 7\}$, ML and MWPM yield statistically indistinguishable logical error rates. For depolarizing noise, where ML is expected to outperform MWPM, any advantage remains below the present statistical resolution. The ML decoder thus serves mainly as a rigorous lower bound and benchmarking reference, though its computational cost makes it impractical for real-time decoding.

I. INTRODUCTION

Quantum error correction (QEC) is indispensable for large-scale quantum computation. Physical qubits are inherently noisy, and without active correction errors accumulate faster than useful computation can proceed. The surface code [1, 2] is the leading candidate for practical fault-tolerant quantum computing owing to its high error threshold ($\sim$1%), geometrically local stabilizer measurements, and planar layout that is amenable to semiconductor and superconducting fabrication.

A critical ingredient in any QEC protocol is the *decoder*: the classical algorithm that processes stabilizer measurement outcomes (the *syndrome*) and infers which correction to apply. Real quantum hardware exhibits *spatially varying* noise [3]: different qubits and gates on the same chip experience different error rates due to fabrication imperfections, material disorder, and environmental inhomogeneity. This noise spectrum can be characterized experimentally and is therefore available to the decoder before operation begins. The central question is whether a decoder that incorporates this site-level noise information can outperform one that assumes a spatially uniform error model.

The minimum-weight perfect matching (MWPM) decoder [4] is the workhorse of current QEC experiments. It can accommodate heterogeneous noise through noise-adapted edge weights, but fundamentally approximates the decoding problem by selecting the single most likely error chain. The maximum likelihood (ML) decoder, by contrast, marginalizes over *all* error configurations consistent with the syndrome and is therefore optimal by construction [5]. An efficient approximation to ML decoding is obtained via tensor-network (TN) contraction [5–7], at the cost of substantially increased classical computation.

In this work, we implement such a TN-ML decoder for the unrotated surface code, integrated with the `stim` stabilizer simulator [8]. We test the decoder on three noise models: a heterogeneous depolarizing channel, a spin-qubit Z-dephasing channel, and an Exchange only (EO) qubit X, Z-dephasing channel. Each captures distinct hardware physics and provides a different test of the potential gain from full noise-model access.

The remainder of this paper is organized as follows. Section II reviews the surface-code QEC pipeline and the two decoders. Section III describes the TNDecoder implementation. Section IV details the three noise models and their physical origins. Section V presents simulation results. Section VI discusses the implications, and Sec. VII concludes with an outlook.

II. BACKGROUND

A. Surface code and the QEC pipeline

The unrotated surface code of distance d encodes one logical qubit into $d^2 + (d-1)^2$ physical qubits arranged on a 2D lattice [1]. The QEC cycle has four stages: (1) quantum information is stored in physical qubits; (2) stabilizer measurements detect the locations of errors and produce the syndrome $s \in \{0, 1\}^m$; (3) a decoder uses s and the noise model to infer the most likely logical coset L to apply; (4) the inferred correction restores the logical state and the cycle repeats. The figure of merit is the *logical error rate*, approximated as

$$p_L \approx 1 - (1 - p_L^x)(1 - p_L^z), \tag{1}$$

where p_L^x and p_L^z are the logical X- and Z-error rates. We study unrotated surface codes with distances $d \in \{3, 5, 7\}$, assuming perfect initialization and noiseless syndrome measurement throughout.

[*] https://github.com/Minyoung-Kim1110/TNDecoder/tree/master

B. Maximum likelihood decoder

Given syndrome s, the ML decoder selects the logical coset with the highest total probability:

$$\hat{L} = \arg\max_{L} \sum_{\substack{E:\,\partial E = s \\ [E]=L}} \Pr(E), \tag{2}$$

where $\Pr(E)$ is computed from the full, site-dependent noise model. This is optimal by definition—no decoder can achieve a lower p_L for a given noise model. Although exact degenerate maximum-likelihood decoding is #P-complete for general stabilizer codes, the specific 2D surface-code problem studied by Ref. [5] is polynomial-time under independent X/Z noises; here we use approximate 2D PEPS contraction to treat more general noise models. (Sec. III).

C. MWPM decoder

The MWPM decoder builds a graph from syndrome defects, assigns edge weights derived from local error probabilities, and returns the minimum-weight perfect matching. It effectively selects the *single* most likely error chain rather than marginalizing over all chains. MWPM is near-linear in qubit count and is implemented here via PyMatching [4] with noise-adapted edge weights. Table I summarizes the two decoders.

TABLE I. Comparison of the MWPM and ML decoders.

Feature	MWPM	ML (this work)
Speed	Fast	Slow
Noise adapt.	Partial (weighted graph)	Full (site-dependent)
Optimality	Approximate	Optimal (in principle)
Implement.	PyMatching	TN-PEPS

III. IMPLEMENTATION

A. TNDecoder package

The ML decoder is implemented in the TNDecoder package [9], written in Python and fully compatible with the `stim` stabilizer simulator [8]. It accepts `stim.DetectorErrorModel` objects and detection-event arrays directly, making it a drop-in replacement for other decoders in existing `stim`-based pipelines without modification of the sampling workflow.

B. Syndrome generation

Stabilizer circuits for the unrotated surface code are compiled and sampled using `stim`. Per-qubit error channels are specified as a dictionary mapping qubit indices to error parameters, enabling spatially heterogeneous noise at no additional overhead.

C. Tensor-network contraction

For each syndrome, the decoder constructs a PEPS tensor network whose partition function encodes the marginal probability of each logical coset. Approximate boundary matrix product state (MPS) contraction is used to keep the computational cost tractable. The bond dimension χ controls the trade-off between approximation quality and runtime.

IV. NOISE MODELS

A. Heterogeneous depolarizing channel

The single-qubit depolarizing channel with error rate p is

$$\mathcal{E}_{\mathrm{dep}}(p)[\rho] = (1-p)\,\rho + \frac{p}{3}\big(X\rho X + Y\rho Y + Z\rho Z\big). \tag{3}$$

Spatial inhomogeneity is modeled by drawing each qubit's error rate p_i from a truncated Gaussian distribution,

$$p_i \sim \mathcal{N}\big(p_0,\,(\sigma p_0)^2\big), \tag{4}$$

where p_0 is the mean error rate and σ is the relative standard deviation. This quasistatic model captures the run-to-run variability observed in superconducting and spin-qubit platforms while remaining analytically tractable.

B. Spin qubit: Z-dephasing channel

A single-spin qubit in a quantum dot is controlled by an applied static magnetic field B_0, with the Zeeman splitting $\Delta E = g\mu_B B_0$ setting the qubit frequency. In micromagnet-based architectures, the local magnetic field experienced by the electron depends on its spatial position inside the confining potential, because the micromagnet produces a fringe field with a finite spatial gradient [10].

$1/f$ charge noise—low-frequency voltage fluctuations arising from fluctuating charge traps in the gate oxide or substrate [11]—perturbs the gate voltages and displaces the electron's confining potential. The electron therefore samples a slightly different position in the micromagnet's fringe field, producing a random fluctuation in the effective Zeeman splitting it experiences. This fluctuation is *longitudinal*: it shifts the energy splitting between $|\uparrow\rangle$ and $|\downarrow\rangle$ without driving transitions between them, resulting in pure dephasing. In the Pauli error picture this is a Z-type channel,

$$\mathcal{E}_Z(p_i^Z)[\rho] = (1-p_i^Z)\,\rho + p_i^Z\, Z\rho Z, \tag{5}$$

with no X- or Y-type component. The rate p_i^Z varies across the device according to Eq. (4), reflecting spatial inhomogeneity in the local charge-trap density.

C. Exchange-only qubit: X,Z-dephasing channel

1. Qubit structure and control

The exchange-only (EO) qubit encodes one logical qubit in the $S_{\text{tot}} = \frac{1}{2}$ decoherence-free subspace of three electron spins confined in a linear triple quantum dot (TQD) [12]. As shown in Fig. 1, the device consists of three quantum dots (spin sites P1, P2, P3) connected by two tunable tunnel barriers (X1 between dots 1 and 2, X2 between dots 2 and 3). The plunger gates P_1, P_2, P_3 control the on-site chemical potentials of each dot, while the barrier gates X_1 and X_2 set the interdot tunnel couplings. Together these five voltages determine the two nearest-neighbor exchange couplings J_{12} and J_{23}, which are the sole control parameters of the logical qubit.

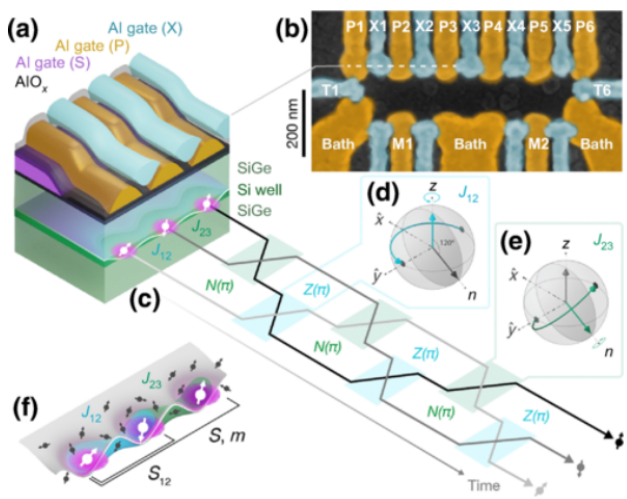

FIG. 1. Schematic of the exchange-only qubit and its control axes adapted from Ref. [13]. Three electron spins occupy a linear triple quantum dot, with plunger gates P_i tuning the dot potentials and exchange gates X_1 and X_2 controlling the couplings J_{12} and J_{23}. In the encoded subspace, pulses on X_1 and X_2 generate rotations about the logical axes $\hat{z}$ and $\hat{n}$, respectively. Charge noise on these gates perturbs the exchange couplings and thereby induces effective X- and Z-type errors.

2. Control axes and the two-rotation structure

In the encoded two-level subspace the effective Hamiltonian is [12, 14]

$$H = -\tfrac{1}{2}\Big[J_{12}\,\sigma_z + J_{23}\big(\frac{1}{2}\sigma_z + \frac{\sqrt{3}}{2}\sigma_x\big)\Big], \tag{6}$$

where σ_x, σ_z are Pauli matrices in the logical basis. Switching on J_{12} alone generates a rotation about $\hat{z}$, while switching on J_{23} alone generates a rotation about the tilted axis $\hat{n} = (-\frac{\sqrt{3}}{2}, 0, -\frac{1}{2})$, which lies 120° from $\hat{z}$ in the x-z plane. An arbitrary single-qubit gate is constructed from a sequence of $\hat{z}$-pulses (using J_{12}) and $\hat{n}$-pulses (using J_{23}).

3. Why charge noise gives X and Z errors

$1/f$ charge noise introduces random voltage fluctuations δV on *all* gates—both plunger gates P_1, P_2, P_3 and barrier gates X_1, X_2. Because the EO qubit is entirely electrically controlled, every such fluctuation feeds directly into one or both of the exchange couplings. Specifically, fluctuations on P_1, P_2, or the two barrier gates primarily shift J_{12}, while fluctuations on P_2, P_3, or the barrier gates primarily shift J_{23}; in practice, perturbations on most gates couple to both couplings simultaneously.

The key consequence follows from the two-axis control structure in Eq. (6): a random perturbation δJ_{12} changes the rotation angle about $\hat{z}$, contributing a Z-type rotation error, while a simultaneous random perturbation δJ_{23} changes the rotation angle about $\hat{n}$, which has a nonzero projection onto σ_x and hence contributes an X-type error. Unlike the spin qubit (Sec. IV B), where charge noise couples to only one axis (the magnetic field gradient axis) and produces purely longitudinal dephasing, in the EO qubit charge noise couples to *both* control axes because the qubit is defined and operated entirely through the charge degree of freedom. A single noise source therefore simultaneously randomizes two non-commuting rotation axes, producing errors with both Z and X components.

In the Pauli error picture this yields a biaxial channel,

$$\mathcal{E}_{XZ}(p_i^X, p_i^Z)[\rho] = (1 - p_i^X - p_i^Z)\,\rho + p_i^X\, X\rho X + p_i^Z\, Z\rho Z, \tag{7}$$

where p_i^X and p_i^Z are drawn independently from the heterogeneous Gaussian model of Eq. (4), reflecting spatial variation in the charge environment across the qubit array.

V. RESULTS

A. Depolarizing channel

Figure 2 shows the logical error rate p_L as a function of mean physical error rate p_{mean} for the heterogeneous depolarizing channel, comparing ML and MWPM at $d = 3, 5, 7$.

The ML and MWPM results are statistically consistent across all distances and physical error rates simulated ($p_{\text{mean}} \in [10^{-2}, 10^{-1}]$), with both showing the expected suppression of p_L when $p < p_{th}$ with increasing code distance. The depolarizing channel is theoretically a case where ML should outperform MWPM: MWPM decodes X- and Z-type errors independently on its syndrome graph and does not account for the correlated structure of the depolarizing channel (in particular, that a Y error is a simultaneous X and Z error). The ML decoder, through the full marginalization in Eq. (2), naturally incorporates

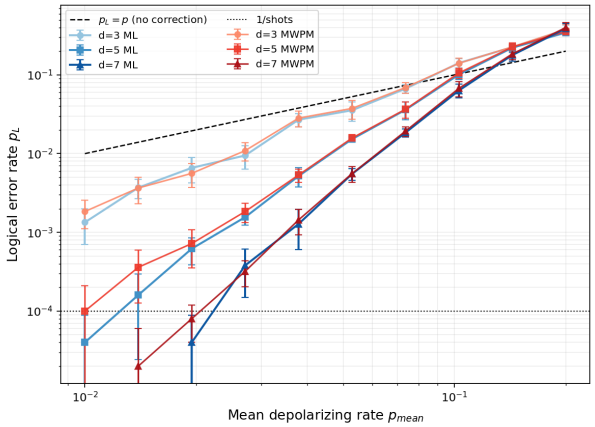

FIG. 2. Logical error rate vs. mean depolarizing rate for ML (blue) and MWPM (red) at $d = 3, 5, 7$. The decoders agree within statistical error bars throughout. The dotted horizontal line marks $p_L = 1/N_{\text{shots}}$; estimates below this line are unreliable.

these correlations. However, the predicted improvement is not visible at the current shot count; differences between ML and MWPM are smaller than the statistical uncertainty set by $1/N_{\text{shots}} = 10^{-4}$. Resolving the improvement would require substantially more shots, particularly at low physical error rates where p_L approaches the shot-count floor.

B. Hardware-informed channels

Figure 3 shows p_L for the two hardware-informed channels.

Under Z-only dephasing noise (spin qubit), the syndrome graph separates cleanly into independent X- and Z-type subgraphs, and noise-adapted MWPM already captures most of the available information through its edge weights, leaving little room for the ML decoder to improve. Under the biaxial X, Z EO noise, the same qualitative picture holds: at these code distances the dominant correction class is well identified by MWPM's graph-based approach, and the full marginalization performed by ML does not yield a statistically distinguishable gain.

VI. DISCUSSION

A. ML as a lower bound on logical error rate

A central role of the ML decoder in this study is to provide a *lower bound* on the logical error rate achievable by any decoder. Because ML selects the logical coset with the highest total probability over all consistent error configurations, no other decoder can achieve a lower p_L for the same noise model. The ML results therefore set a performance ceiling: any gap between the MWPM

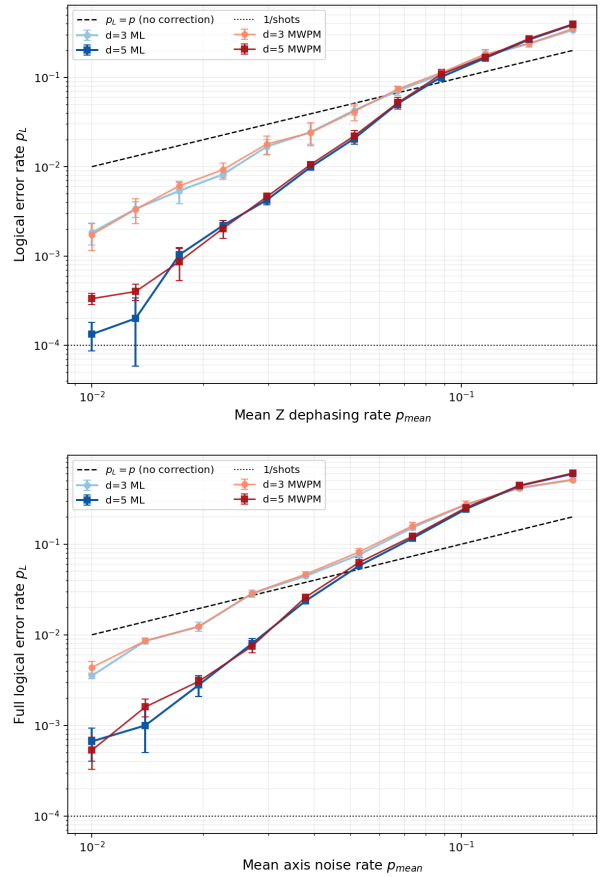

FIG. 3. Logical error rate under hardware-informed noise. Top: spin-qubit Z-dephasing channel. Bottom: EO-qubit X, Z-dephasing channel. ML and MWPM curves are within statistical error bars in both cases.

curve and the ML curve is the maximum possible gain from algorithmic improvement. In our experiments this gap is consistent with zero within statistical uncertainty, indicating that noise-adapted MWPM is already near-optimal for the noise models and code sizes studied here.

B. Statistical limitations

The comparison is fundamentally limited by shot count. Error bars become large once p_L falls below $1/N_{\text{shots}}$ because fewer than one logical error is expected per shot on average, placing estimates in that regime under heavy Poisson fluctuation. For the depolarizing channel, the theoretically expected advantage of ML over MWPM may exist but is smaller than the statistical noise at current shot counts. Definitively resolving this would require either a substantially larger number of shots (scaling as $1/\Delta p_L^2$, where Δp_L is the expected improvement) or a targeted experiment at intermediate physical error rates where the predicted advantage is largest.

C. Computational cost

The ML decoder is computationally heavy due to the cost of PEPS boundary-MPS contraction, which scales exponentially in the bond dimension χ and polynomially in the lattice size. For $d = 7$, each decode call is orders of magnitude slower than the corresponding MWPM call. This makes the current implementation unsuitable for real-time decoding, where the decoder must keep pace with the physical gate clock ($\sim \mu$s timescales in most platforms). The decoder is therefore best understood as a benchmark tool: a theoretically optimal reference against which faster, approximate decoders can be calibrated.

VII. CONCLUSION

We have implemented, validated, and benchmarked a noise-adapted tensor-network maximum likelihood decoder for the surface code, arriving at the following conclusions.

(i) *Implementation.* A fully functional ML decoder compatible with the `stim` stabilizer simulator has been implemented in the TNDecoder package. It accepts per-qubit, per-channel noise specifications and integrates seamlessly into existing `stim`-based pipelines.

(ii) *Testing under heterogeneous noise.* The decoder has been tested on three noise models: heterogeneous depolarizing noise, spin-qubit Z-dephasing, and EO-qubit X, Z-dephasing, for code distances $d \in \{3, 5, 7\}$.

(iii) *Performance vs. MWPM.* Under all noise models tested, ML and MWPM deliver statistically indistinguishable logical error rates. For the depolarizing channel, ML is theoretically expected to outperform MWPM due to correlated Y-error structure, but the improvement is smaller than statistical fluctuations at the shot counts used.

(iv) *Lower bound.* The ML decoder establishes a rigorous lower bound on the achievable logical error rate. In the regimes studied this bound is consistent with the MWPM result, confirming that noise-adapted MWPM operates near the optimal limit at these code distances.

(v) *Computational cost.* The ML decoder is computationally heavy and is best used as a benchmarking and analysis tool rather than a real-time decoder.

Looking ahead, several natural extensions remain: extending the decoder to handle space-time (3D) syndrome data, consider noise correlations, imperfect initialization syndrome extraction; modeling spatially and temporally correlated $1/f$ noise beyond the quasistatic approximation to better capture slow temporal fluctuations in charge-trap environments; and relaxing the assumptions of perfect initialization and noiseless syndrome measurement to enable full circuit-level noise simulation, the practically relevant regime for fault-tolerant operation.

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